Short-PHD: Detecting Short LLM-generated Text with Topological Data Analysis After Off-topic Content Insertion

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Abstract

The malicious usage of large language models (LLMs) has motivated the detection of LLM-generated texts. Previous work in topological data analysis shows that the persistent homology dimension (PHD) of text embeddings can serve as a more robust and promising score than other zero-shot methods. However, effectively detecting short LLM-generated texts remains a challenge. This paper presents Short-PHD, a zero-shot LLM-generated text detection method tailored for short texts. Short-PHD stabilizes the estimation of the previous PHD method for short texts by inserting off-topic content before the given input text and identifies LLM-generated text based on an established detection threshold. Experimental results on both public and generated datasets demonstrate that Short-PHD outperforms existing zero-shot methods in short LLM-generated text detection. Implementation codes are available online.[1]

1 Introduction

Table 1: Detection performance (AUC) of PHD and Short-PHD for identifying short LLM-generated texts (50 tokens). Results are averaged from Table 4.

Method	OPT/OPT-iml/GPT-NeoX	GPT-3.5/GPT-4o
PHD (Tulchinskii et al., 2024)	0.632	0.653
Short-PHD (Ours)	**0.793** **(Improve ↑ 25.47%)**	**0.830** **(Improve ↑ 27.11%)**

With the rapid development of Large Language Models (LLMs) (Guo et al., 2025), detecting LLM-generated content has become a crucial issue in controlling malicious content on the internet (Lin et al., 2022; Liu et al., 2024; Fang et al., 2024). In this work, we focus on zero-shot methods for detecting LLM-generated texts. A typical zero-shot method leverages an LLM as a scoring model to compute the conditional probability of each token given previous tokens and analyzes the statistical properties of these probabilities (Ippolito et al., 2020; Su et al., 2023; Mitchell et al., 2023). For example, DetectGPT (Mitchell et al., 2023) and Fast-DetectGPT (Bao et al., 2024) focus on conditional probability curvature, which evaluates the direction of perplexity change after perturbing random tokens. However, to ensure reliable performance, these methods often require a 'white-box' setting in which the generative LLM can serve as the scoring model during detection. For example, DetectGPT achieves a detection AUC of only 0.6 when using GPT-J as the scoring model to detect GPT-2-generated text (Mitchell et al., 2023).

To overcome the 'white-box' challenge, apart from probability-based methods, Tulchinskii et al. (2024) propose utilizing the Persistent Homology Dimension (PHD) from topological

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[1]https://github.com/djwei96/ShortPHD

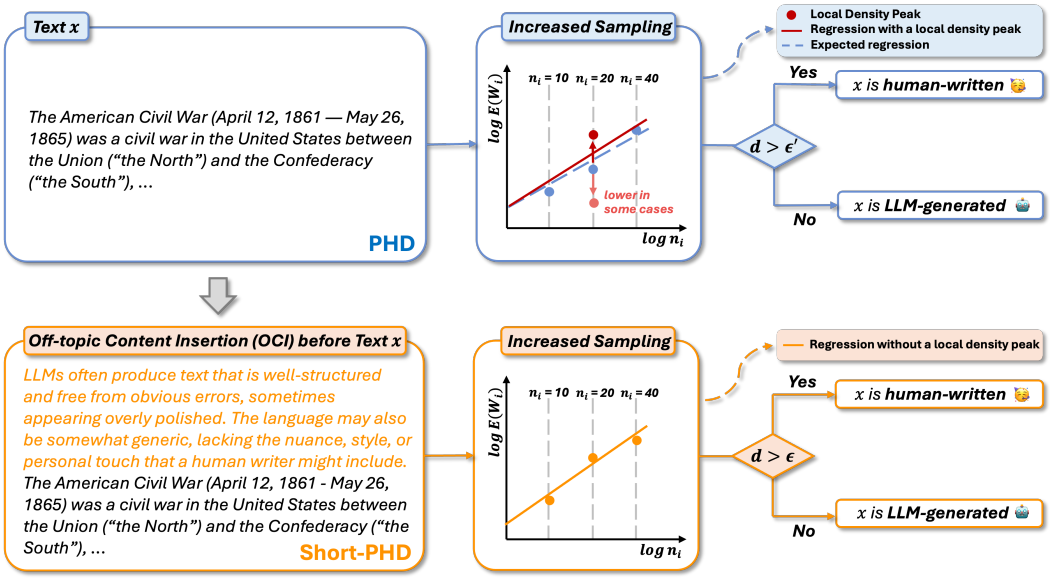

Figure 1: PHD (Tulchinskii et al., 2024) v.s. Short-PHD. n_i is the number of sampled points in the point cloud W. W_i is the sampled subset of points in W with size n_i. $E(W_i)$ is computed by the minimal spanning tree (MST).

data analysis to identify LLM-generated texts. Specifically, texts are first embedded using a pretrained LLM such as RoBERTa (Liu et al., 2019). Next, the PHD (Kališnik, 2019), a concept of topological data analysis (Chazal & Michel, 2021), is computed for these embeddings. From a topological perspective, PHD measures the connectedness of the embedding, and a higher PHD represents a less connected data structure. **LLM-generated texts are found to exhibit lower PHD values than human-written text** due to their naturally higher connectedness in the embedding space. PHD outperforms other zero-shot methods, such as DetectGPT, in detecting LLM-generated texts across various models and domains, and also demonstrates greater robustness against adversarial attacks (Tulchinskii et al., 2024).

However, similar to other zero-shot methods, the PHD method can only detect text that is sufficiently long, and Tulchinskii et al. (2024) assume LLMs are fluent enough to generate at least 200 tokens, which limits their applicability for detecting short texts. In this study, we first analyze the scalability challenges of the current PHD method. Specifically, based on the increased sampling technique used to estimate PHDs, the PHD method encounters a phenomenon named local density peaks, which can make estimation unstable (either too high or too low). These local density peaks are particularly problematic for short texts, as they lack sufficient variability to sample out of the peak.

To facilitate the PHD method to overcome the local density peak issue in short texts, we propose a hypothesis involving an off-topic content insertion (OCI) technique. Specifically, we hypothesize that **inserting off-topic content unrelated to the input text can stabilize and increase the PHD score**. The underlying rationale is that by adding extra content, the text becomes longer, helping the PHD estimation avoid local density peaks in increased sampling. Furthermore, since PHD measures the connectedness of embeddings, limiting the inserted content to off-topic (i.e., irrelevant) content ensures that the PHD increases for all inputs because text embeddings are less connected.

Based on OCI, we present Short-PHD, a zero-shot method derived from the original PHD method, specifically tailored to detect short LLM-generated texts. A comparison between Short-PHD and PHD is illustrated in Figure 1. Specifically, given a short human-written text and a short LLM-generated text, Short-PHD first inserts the same off-topic content into each text. Then, it applies increased sampling to compute their respective PHD scores. According to the hypothesis on OCI, introducing OCI stabilizes and increases the PHD values of both

human-written and LLM-generated texts. To preserve the effectiveness of detection, we further hypothesize that **the gap between human-written and LLM-generated PHD values remains the same in expectation after inserting the same off-topic content**. Consequently, Short-PHD can establish a threshold of PHD to identify LLM-generated texts.

Contributions. We summarize our main contributions as follows: 1) We identify the local density problem in estimating PHDs of short texts in the previous PHD method and propose that OCI could alleviate this issue. 2) We propose a novel Short-PHD method based on OCI to detect short LLM-generated texts. 3) Experimental results on both public and generated datasets indicate that Short-PHD achieves state-of-the-art performance in detecting short LLM-generated texts compared to the previous PHD method and other baselines.

2 Preliminary

The persistent homology dimension (PHD) analyzes the data in high dimensional space about the birth and death of its persistence cycles (Tulchinskii et al., 2024). Empirical evidence shows that LLM-generated texts exhibit lower PHD values than human-written texts. Formally, let $\mathcal{W} \subset \mathbb{R}^d$ be a d-dimensional topological space, and let $W = \{w_i \in \mathbb{R}^d\}$ be a discrete point cloud sampled from $\mathcal{W}$, i.e., $W \subset \mathcal{W} \subset \mathbb{R}^d$. In our work, we represent text as a point cloud in a high-dimensional space using embeddings from a pretrained LLM such as LLaMA (Dubey et al., 2024). Specifically, given a text sequence of n tokens, each token is mapped to a d-dimensional embedding vector, forming a set of n points in $\mathbb{R}^d$. For example, with $d = 768$, the text forms a point cloud in $\mathbb{R}^{768}$, where each point corresponds to a token embedding.

2.1 Mathematics Preliminary

We provide basic concepts of homology groups in Appendix A.1. Informally speaking, the idea of homology groups is to find non-trivial topological features in a topological space. These features include connected components, holes, and cavities that exist in the space. Next, we introduce the mathematical concepts of persistent homology in Appendix A.2. To sum up, different from homology groups that analyze a topological space, persistent homology belongs to data analysis that analyzes data points (or a point cloud). First, from the point cloud W, the Vietoris-Rips (VR) complexes $\mathrm{VR}(W)$ (Bakhtin et al., 2019) are constructed with a filtrating parameter r. By changing the filtrating parameter r, persistent homology focuses on the birth and death of cycles $(t_{\text{birth}}, t_{\text{death}})$ in the complex. **The persistent homology dimension (PHD) is defined based on the lifetime of these cycles.** We refer readers to Appendix A.2 and previous work (Birdal et al., 2021) for more details.

2.2 Estimating Persistence Homology Dimension

Following previous work (Tulchinskii et al., 2024), we focus on PH_0, which measures the 0-persistent homology. The PH_0 corresponds to the minimal spanning tree (Tulchinskii et al., 2024) and Equation A1 (in Appendix A.2) can be calculated as

$$E_\alpha^0(W) = \sum_{e \in \mathrm{MST}(W)} |e|^\alpha, \tag{1}$$

where $\mathrm{MST}(W)$ denotes the minimal spanning tree (MST) from point set W, e is the edge in the tree, and $|e|$ is the length of e. The exact number of PH_0 is hard to compute even for a small point cloud, but it could be approximated through increased sampling. The increased sampling is conducted by sampling different numbers of data points in the point cloud and estimating their MST. Specifically, if points w_i in a cloud are independently distributed, then with probability approaching one, $E_\alpha^0(W) \sim Cn^{\frac{d-\alpha}{d}}$ holds as $n \to \infty$ (Steele, 1988). Therefore, in practice, by increasing the number n_i of sampled points in a subset $W_i = \{w_1, w_2, \ldots, w_{n_i}\}$, we can find its minimal spanning tree and compute $E_\alpha^0(W_i)$. Since we know that $\log E_\alpha^0(W_i) \sim \left(1 - \frac{\alpha}{d}\right) \log n_i + \log C$, we can fit a linear regression to compute